Towards ML Modeling and Deployment: Optimizing Fitting Time, Efficiency and Productivity using MLflow

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MOTIVATION

After building AI/ML models, what next?

Transitioning models from development to production can be challenging; VentureBeat notes 87% of ML projects never reach this stage. By leveraging requisite MLOps tools and frameworks such as Apache MLflow and WandB, the ML lifecycle can be efficiently managed through experiment tracking, model registry, and model deployment. This will ensure ML models deliver their intended value with minimum time as possible. The project provides an end-to-end walkthrough using **Apache MLflow**.

METHODOLOGY

MODEL TRACKING – Training + Hyperparameter Tuning

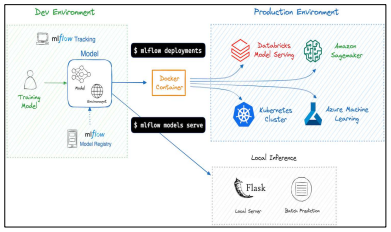
Linear regression model was used over alpha and L1_ratio values—Ridge, Lasso or Elastic Net (**MLflow UI & Logging**)

MODEL REGISTRY – UI and API Workflows

Packages the model and its dependencies for productionization. Facilitates model lineage, versioning, aliasing, tagging, and annotations (**MLflow Artifacts).**

MODEL TO PRODUCTION – Kubernetes Clusters

MLflow offers easy toolset for deploying ML models to various targets, including local environments, cloud services, and Kubernetes clusters.



Deployment Targets

databricks

Microsoft Azure

Amazon SageMaker

kubernetes

SAMPLE DATASET

Sample dataset used for the experiment is the **wine quality** dataset from from the UCI Machine Learning Repository—13 features | 178 samples. Label: quality (0 to 2).



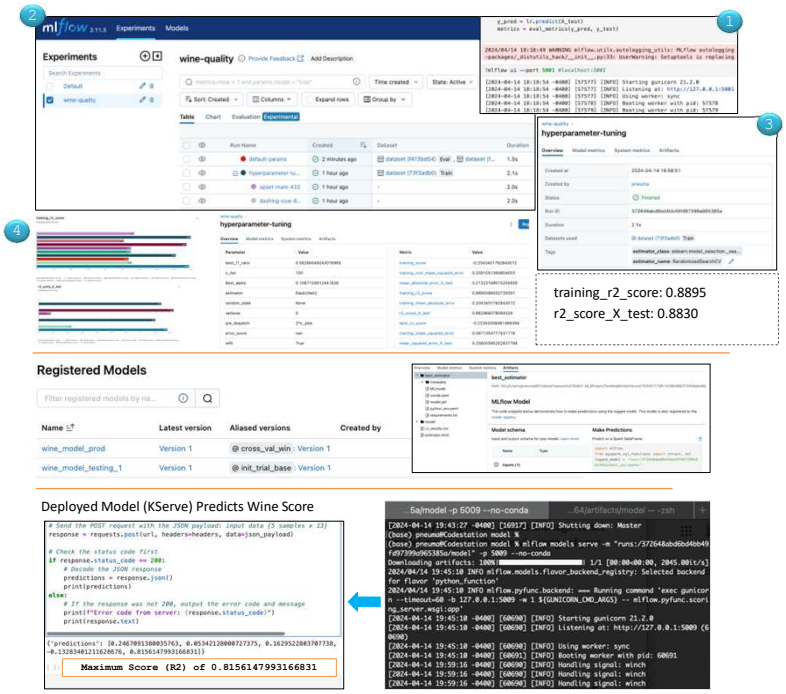
Data Features

- alcohol
- malic_acid
- ash
- alcalinity_of_ash
- magnesium
- total_phenols
- flavonoids
- nonflavanoid_phenols
- proanthocyanins
- color_intensity
- hue
- od280/od315_of_diluted_wines
- proline

MLOPS LIFECYCLE & TOOLING

- Mlflow
- Kind (Kubernetes in Docker)
- MLflow Auto-logging
- Minikub
- KServe
- MLServer
- Flask
- Apache Spark

RESULTS



CONCLUSION & FUTURE WORK

From the experiment and result shown, MLflow provides simplified framework to track ML models, register models and deploy to production using Kubernetes clusters.

The model "wine_model_prod" was successfully implemented using MLflow to accurately predict the wine quality with R2 score of 88.95% on training set and 88.3% on testing set.

REFERENCES

Bishop, C. M. (2007). Pattern Recognition and Machine Learning. Springer. ISBN 978-0387310732

Understanding ETL by O'Reilly: https://www.databricks.com/

MLflow: https://mlflow.org/

Kubeflow vs Mlflow: https://www.youtube.com/watch?v=eZlKCY8QzNU

Acknowledgements

We extend our deepest gratitude to Dr. Evans Lucas for his essential contributions to our project. His expert guidance and profound insights have been fundamental in shaping our research and enhancing the integrity of our findings. Dr. Lucas's dedication to high standards and his generous sharing of knowledge have significantly enriched our project's journey. We deeply appreciate his steadfast support and commitment throughout this process.

RESULTS

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